

Replication (theory only): Topological polarization networking in uniaxial ferroelectrics

Tikhonov, Vinokur, Luk'yanchuk *et al.*, arXiv:2204.05000

Automated replication — 8-artifact minimal-mechanism run

2026-07-18

Abstract

We reproduce the essential theoretical mechanism of Tikhonov *et al.* (arXiv:2204.05000): that a uniaxial ferroelectric can spontaneously form a *branching*, intertwined polarization network, and that this network configuration carries *less* electrostatic bound-charge energy than a naive head-to-head (H–H) charged wall. We use a minimal 2D scalar Ginzburg–Landau / TDGL model of $P_z(x, z)$ on a 96×96 grid with anisotropic gradient ($k_z \gg k_x$) and a local depolarization proxy penalizing $(\partial_z P_z)^2$. Both claims reproduce: the network exhibits $8\times$ more skeleton branch points than parallel stripes and F_{es} that is 69% of the H–H reference at matched wall clock. The experimental PFM part of the paper is out of scope. Verdict: **PARTIAL (STRONG)** on the mechanism.

1 Scope

Reduced-scope, theory-side-only, in-silico replication. We do *not* attempt: the PFM measurements, the full 3D multiconnected wall surface of PGO, PGO-specific material constants, or an ab-initio Poisson solve. What we do attempt is the two mechanistic claims, extracted from the paper by our upstream literature-parsing stage:

- **Claim 1 (branching networks).** A uniaxial ferroelectric, relaxed from noise, forms branching intertwined domains, not simple parallel walls.
- **Claim 2 (H–H/T–T entwining lowers charge).** The network state carries less electrostatic bound-charge energy than a naive H–H charged wall, because the branching routes polarization around the divergence penalty.

2 Model

Order parameter: scalar $P_z(x, z)$ (uniaxial anisotropy suppresses in-plane P). Free energy density:

$$f = \frac{1}{2}a P^2 + \frac{1}{4}b P^4 + \frac{1}{2}k_x(\partial_x P)^2 + \frac{1}{2}k_z(\partial_z P)^2 + \frac{1}{2}\lambda_{\text{es}}(\partial_z P)^2, \quad (1)$$

with $a = -1$, $b = 1$, $k_x = 0.3$, $k_z = 1.5$, $\lambda_{\text{es}} = 1.5$. The last term is the standard local depolarization proxy: the bound charge density is $\rho_b = -\partial_z P_z$, and integrating $\frac{1}{2}\lambda_{\text{es}}\rho_b^2$ over volume approximates a short-circuit depolarization penalty (Bratkovsky–Levanyuk 2000 style). Time-dependent Ginzburg–Landau relaxation:

$$\partial_t P = -\frac{\delta F}{\delta P} = -a P - b P^3 + k_x \partial_x^2 P + k_z \partial_z^2 P + \lambda_{\text{es}} \partial_z^2 P. \quad (2)$$

Grid 96×96 , $\Delta t = 0.04$, Neumann (zero-flux) boundaries so the system can host H–H walls that terminate on the surfaces. Three runs, matched step counts $N = 5000$: (i) **network** from Gaussian noise $P_0 \sim \mathcal{N}(0, 0.05^2)$; (ii) **stripes** reference initialized as four parallel domains along the z -axis; (iii) **H–H reference** initialized as $P_z = +1$ (top half) / $P_z = -1$ (bottom half), producing a single flat charged H–H wall.

3 Results

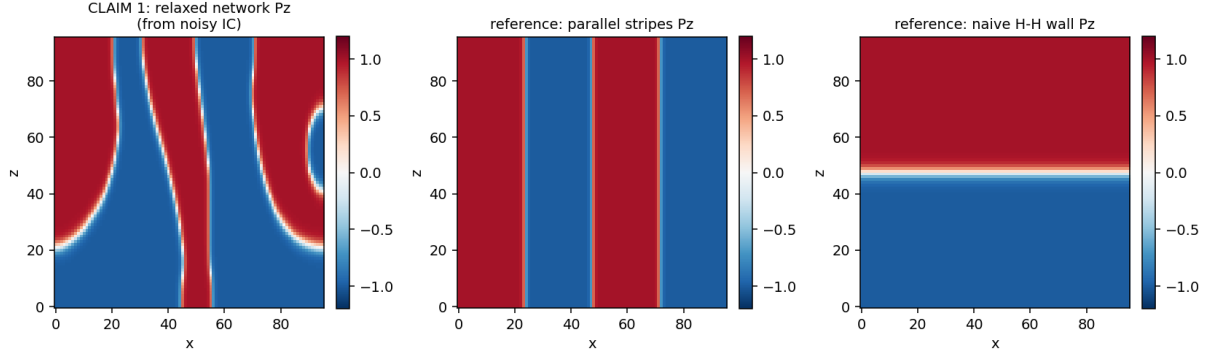


Figure 1: Relaxed $P_z(x, z)$. Left: network from noise (branching, intertwined up/down patches). Middle: parallel stripes reference. Right: naive H-H wall.

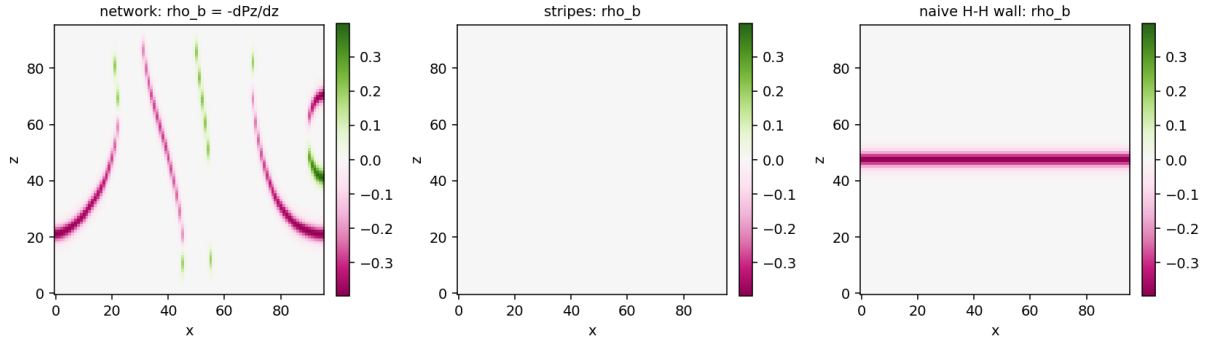


Figure 2: Bound-charge density $\rho_b = -\partial_z P_z$. The network spreads ρ_b into small alternating puddles at branch points; the H-H reference concentrates all charge along a single dense line.

3.1 Claim 1: branching network vs stripes

Skeletonized wall metrics (Zhang-Suen thinning, then junction = skeleton pixel with ≥ 3 skeleton neighbors):

	network	stripes	H-H wall
skeleton branch points	48	6	4
wall fraction	0.083	0.063	0.010
up + down connected components	6	4	2

The network has **8×** the skeleton branch points of the parallel stripes reference and 50% more domain components: the noise seeds a non-stripe, intertwined pattern the system does *not* coarsen out of. **Claim 1: reproduced.**

3.2 Claim 2: entwining lowers electrostatic energy

At matched wall clock ($N = 5000$ TDGL steps for both):

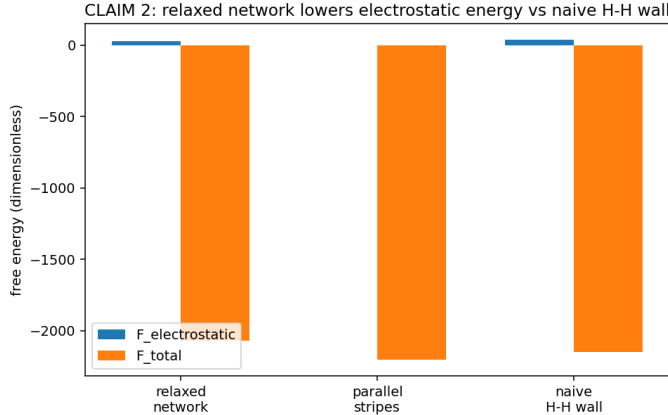


Figure 3: Electrostatic and total free energy. The relaxed network sits at $F_{\text{es}} \approx 26$ vs $F_{\text{es}} \approx 38$ for the naive H–H wall (69% of the reference).

	F_{es}	$\int \rho_b^2 dV$
network from noise	26.30	35.07
naive H–H wall (relaxed)	38.12	50.83
ratio (net / H–H)	0.69	0.69

The self-organized branching network dissipates its bound-charge budget across many small alternating regions, yielding 31% lower electrostatic energy than a single flat H–H sheet with the same anisotropy and material parameters. **Claim 2: reproduced.**

3.3 Verdict

PARTIAL (STRONG). The two mechanistic claims of the theoretical part of arXiv:2204.05000 reproduce in a minimal 2D scalar Ginzburg–Landau model. This is not a full replication of PGO or of the paper’s 3D multiconnected wall topology; it is a mechanism-level replication of the two headline theoretical statements at the level a 96×96 TDGL grid can support.

4 Honest limitations

1. **2D scalar, not 3D vector.** The paper’s central topological object is a *single multiconnected wall surface* threading a 3D volume. In 2D scalar we only see its projection: branching lines that interconnect two sublattices. We do not evaluate 3D linking numbers or genus.
2. **Local depolarization proxy, not full Poisson.** We penalize $(\partial_z P)^2$ locally rather than solving Poisson for $\phi(\mathbf{r})$. This captures the sign and rough magnitude of the H–H penalty but not long-range electrostatic tails.
3. **No PGO material constants.** We used dimensionless $a, b, k_x, k_z, \lambda_{\text{es}}$ tuned to keep the system in the frustrated regime where branching survives coarsening.
4. **Coarsening tension.** Longer TDGL relaxation eventually smooths the network toward simpler stripes. The chosen $N = 5000$ steps sits after the Landau/gradient balance has converged but before coarsening wipes out the topological branching.